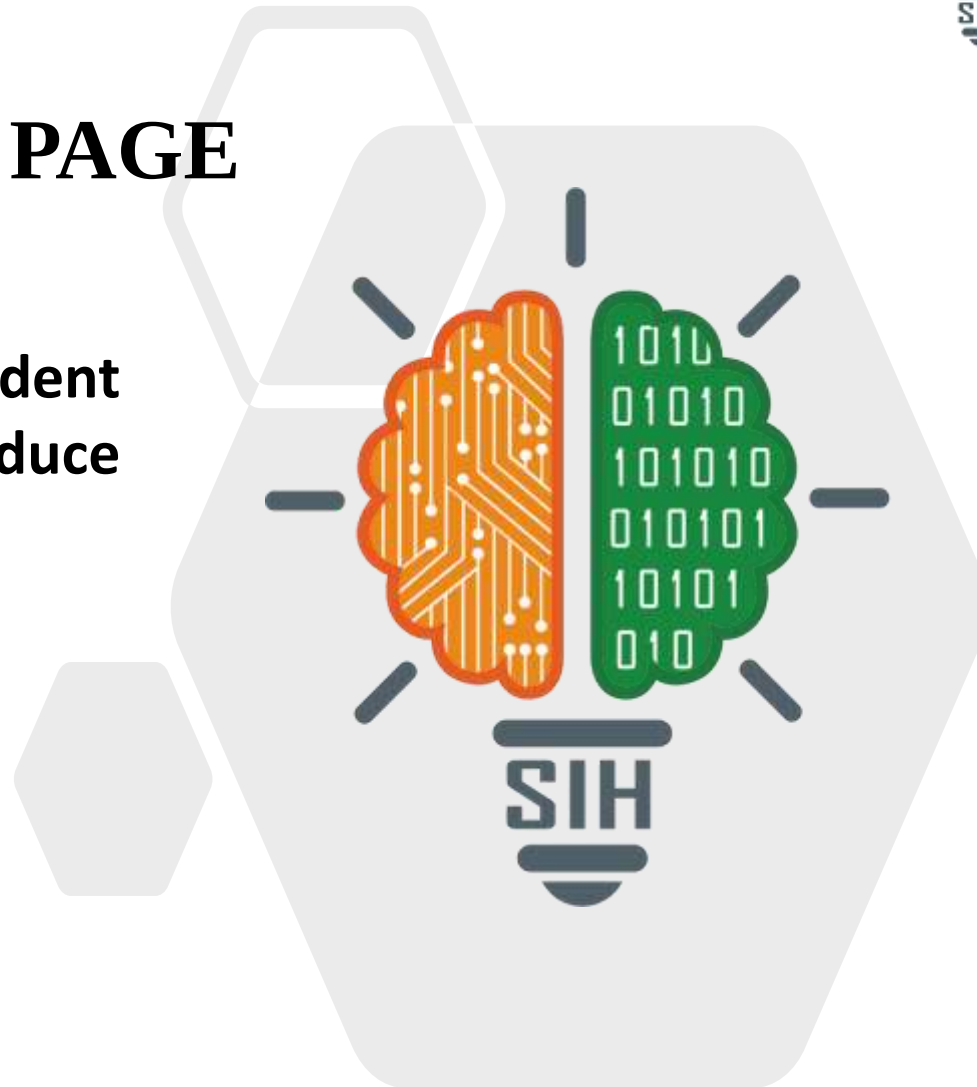


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – SIH26210
- Problem Statement Title- Student
Innovation Agriculture produce
management & processing
- Theme- Agriculture
- PS Category- Hardware
- Team ID-
- Team Name – AgroNova



Student Innovation Agriculture produce management & processing

Idea / Solution

- Digital management of farm produce
- Quality & quantity tracking
- Smart storage recommendations
- Processing of excess/low-grade produce



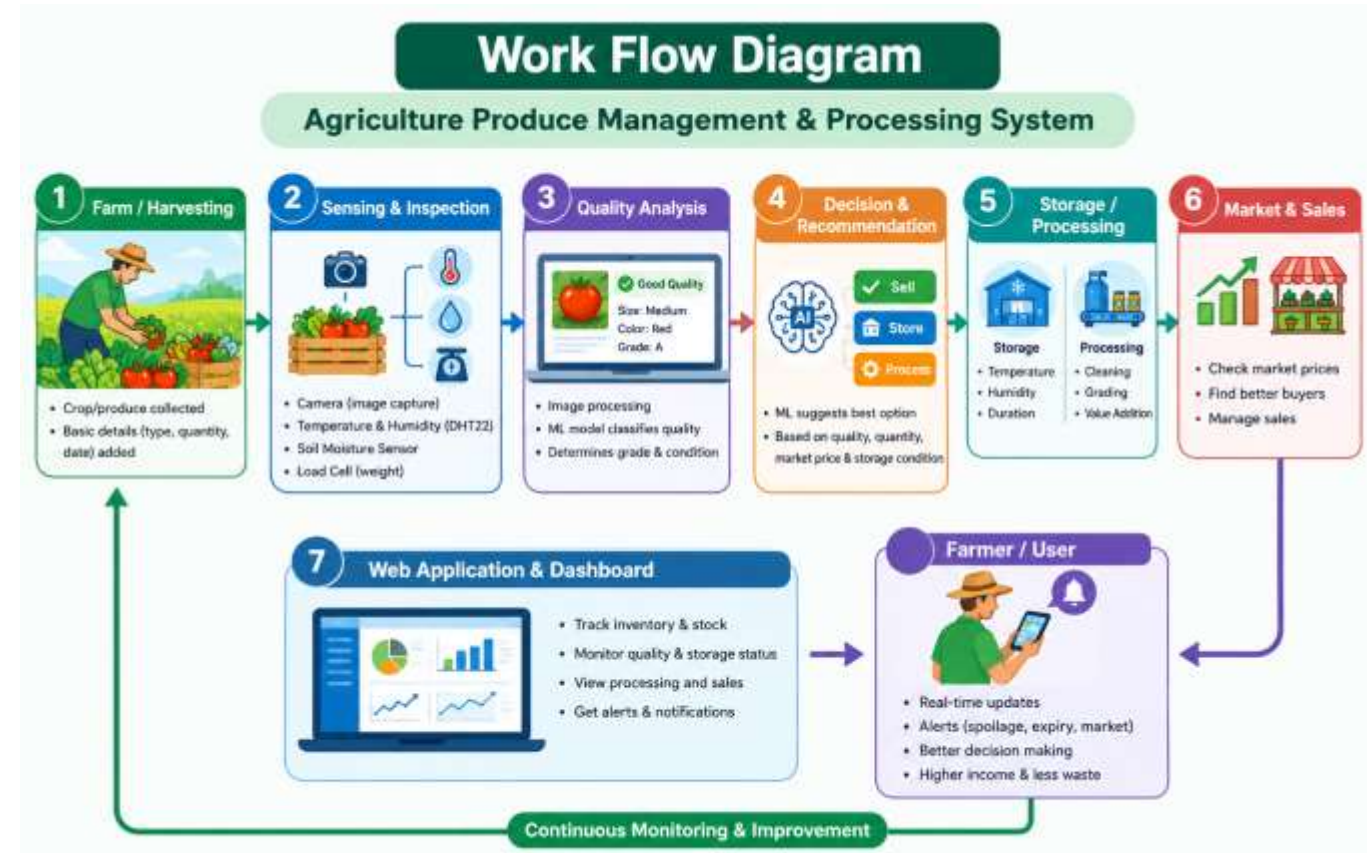
Problem Resolution

- Post-harvest losses → reduce spoilage
- Poor storage → proper storage guidance
- Low-value produce → value-added processing
- Manual records → automated reporting

Unique Value Proposition:

- Farm-to-market management
- Waste-to-value processing
- Quality-based decisions
- Scalable & farmer-friendly system

- **ESP32 / Arduino** – Main control unit
- **Temperature & Humidity Sensor (DHT22)** – Monitors storage conditions
- **Soil Moisture Sensor** – Monitors crop/produce moisture
- **Load Cell + HX711** – Measures produce quantity/weight
- **LCD/OLED Display** – Displays sensor readings and status
- **Buzzer/LED** – Alerts for spoilage or abnormal conditions
- **Wi-Fi Module** – Sends data to the web application
- **Power Supply** – Provides power to the system / 5v or charging model



Technical Feasibility:.

- Uses **IoT, ML & web technologies.**
- Low-cost **ESP32 + sensors + camera.**
- Easy hardware–software integration.

Operational Feasibility:

- **Simple farmer-friendly interface.**
- Easy monitoring of **quality & storage.**
- Automated alerts and reports.

Economic Viability:

- **Reduces post-harvest losses.**
- Increases **farmer income.**
- Low-cost and **scalable solution.**

 Prototype Bill of Materials (BOM)

Component	Purpose	Cost (INR)
ESP32 Microcontroller	Wi-Fi Processing & Logic	₹380
DHT22 Sensor	Temperature & Humidity	₹180
MQ-135 Gas Sensor	Spoilage / Ethylene Gas	₹120
16x2 I2C LCD + Buzzer + LEDs	Local Alert & Visuals	₹180
Box, Power Adapter, Wires	Enclosure & Power	₹340
Total Prototype Cost	₹1,200 – ₹1,850	

Impact

- Reduces post-harvest losses
- Improves produce quality and storage
- Promotes value-added processing
- Supports smart farming decisions
- Reduces food wastage

Benefits

- Increases farmer income
- Reduces storage and processing costs
- Provides better market opportunities
- Saves time and resources
- Scalable for different crops and farmers

Research

- Post-harvest losses are a major challenge in agriculture; globally, **13.3% of food was lost in 2023** between harvest and retail.
- Fruits and vegetables have particularly high losses because of their **high perishability and storage requirements**.

References

- **FAO – Global Food Losses, SDG Indicator 12.3.1**
<https://www.fao.org/sustainable-development-goals-data-portal/data/indicators/1231-global-food-losses/en>
- **ICAR – Post-Harvest Loss & Agro-Processing Research**
<https://icar.gov.in>
- **IoT & ML for Fruit Ripeness and Spoilage Detection (2025)**
IoT and ML-driven platform for fruit ripeness evaluation and spoilage detection.